Python Quiz 4 - File Practice

Instructions:

- Trace each program carefully.
- Write the exact output or exact file contents requested.
- Assume that all files exist and contain exactly the data shown, with no blank lines.

Question 1 - Writing File Contents

Assume that `groups.txt` initially contains exactly:

```
ARCHIVE
```

There is no newline after ARCHIVE.

What will `groups.txt` contain after the following code is executed?

```
groups = ["Art", "Music", "Drama"]

with open("groups.txt", "w") as my_file:
    for group in groups:
        my_file.write(group[0] + str(len(group)))
```

Final contents of `groups.txt`:

```

```

Question 2 - Reading a Header and Columns

The file `destinations.txt` contains:

```
PLACE!ZONE
Museum!C
Library!N
Stadium!E
```

What is displayed by the following code?

```
with open("destinations.txt", "r") as my_file:
    title = my_file.readline()
    code = ""

    for line in my_file:
        line = line.rstrip("\n")
        columns = line.split("!")
        code += columns[0][0] + columns[1]

print(title[0], code)
```

Expected output:

```

```

Question 3 - Processing Words from a File

The file `words.txt` contains:

```
WORD
SILVER
GOLD
BLUE
```

What is displayed by the following code?

```
result = ""

with open("words.txt", "r") as my_file:
    my_file.readline()

    for line in my_file:
        line = line.rstrip("\n")

        if len(line) > 4:
            result += line[-2]
        else:
            result += line[0]

print(result)
```

Expected output:

```

```

Question 4 - Comparing Integer Columns

The file `deliveries.txt` contains:

```
DRIVER::PLANNED::COMPLETED
Ari::8::6
Bo::5::5
Chen::7::9
Devi::6::4
```

What is displayed by the following code?

```
drivers = ""
extra = 0

with open("deliveries.txt", "r") as my_file:
    my_file.readline()

    for line in my_file:
        line = line.rstrip("\n")
        columns = line.split("::")

        planned = int(columns[1])
        completed = int(columns[2])

        if completed >= planned:
            drivers += columns[0][0]
            extra += completed - planned

print(drivers, extra)
```

Expected output:

```

```

Question 5 - Processing Decimal Values

The file `supplies.txt` contains:

```
ITEM$LIMIT$USED
Paint$5.5$4.0
Clay$8.0$8.5
Fabric$6.5$2.5
Ink$3.0$3.0
```

What is displayed by the following code?

```
items = ""
remaining = 0.0

with open("supplies.txt", "r") as my_file:
    my_file.readline()

    for line in my_file:
        line = line.rstrip("\n")
        columns = line.split("$")

        limit = float(columns[1])
        used = float(columns[2])

        if used < limit:
            items += columns[0][0]
            remaining += limit - used

print(items, remaining)
```

Exact output:

Question 6 - Selecting Records Using a Text Column

The file `tasks.txt` contains:

```
TASK@STATUS@HOURS
Setup@DONE@2
Testing@OPEN@3
Review@DONE@1
Report@OPEN@4
```

What is displayed by the following code?

```
result = ""
total_hours = 0

with open("tasks.txt", "r") as my_file:
    my_file.readline()

    for line in my_file:
        line = line.rstrip("\n")
        columns = line.split("@")

        if columns[1] == "OPEN":
            result += columns[0][0]
            total_hours += int(columns[2])

print(result, total_hours)
```

Exact output:

Question 7 - Reading and Writing Files

The file `plants.txt` contains:

```
PLANT#HEIGHT
Fern#18
Palm#35
Moss#7
Rose#22
```

Assume that `selected_plants.txt` initially contains exactly:

```
OLD DATA
```

What will selected_plants.txt contain after the following code is executed?

```
selected = []

with open("plants.txt", "r") as input_file:
    input_file.readline()

    for line in input_file:
        line = line.rstrip("\n")
        columns = line.split("#")

        if int(columns[1]) >= 20:
            selected.append(columns[0])

with open("selected_plants.txt", "w") as output_file:
    for plant in selected:
        output_file.write(plant + "\n")
```

Final contents of `selected_plants.txt`: